

Sewon Kim

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OBJECTIVE

Optimal Control and Robot Learning. My goal is to build a broad-sense control framework for robotic systems that exceeds biological behavior, inspired and enhanced by model-based and learning-based methods.

EDUCATION

University of Seoul

B.E. in Electrical and Computer Engineering, Minor in Mathematics

Advisors: [Gyunghoon Park](#) (Control and Dynamic System Lab), [Dohyun Kwon](#)

Mar 2022 – Aug 2027 (Expected)

Seoul, South Korea

INDUSTRY EXPERIENCE

Drone Researcher at Institute of Innovation for Future Army, Republic of Korea Army

Jan 2025 – Jul 2026

Daejeon, South Korea

AI Lead Engineer at InBody AI Scale Team, InBody

Jan 2024 – Jan 2025

Seoul, South Korea

PROJECTS

• Autonomous air purifying robot agent

4th Place (200 teams) — SK NAMUH X A1: Autonomous Driving Path Optimization (\$2,200)

Independently implemented full-stack autonomous navigation for indoor air purifying robots purely in NumPy, without using open-source code, covering MCL, MPPI, and Traveling Salesman Problem.

Aug 2025 – Sep 2026

• Autonomous racing "F1tenth"

5th Place (12 teams) — 22nd F1tenth Autonomous Grand Prix at CDC Milan 2024

Led the project with 2 undergrads and implemented a full-stack physical autonomous navigation system for racing completely from scratch, covering SLAM, MCL, Trajectory Opt, MPPI.

Mar 2024 – Dec 2024

• Cart pole system design and control by PID and NMPC

3rd Place (25 teams) — UOS ECE Innovation Fair (\$220)

Independently developed a physical Cart-Pole system for \$272, covering modeling, motor control, sensor integration, state estimation, embedded systems, and MPC.

Jul 2024 – Sep 2024

• InBody AI Scale

CES Las Vegas 2025

Developed height estimation model (accuracy 99.4%) and face recognition system of "InBody AI Scale", fully automated body composition analyzer that reduces measurement time from 2 minutes to 40 seconds.

Jan 2024 – Jan 2025

• ECG-based cardiovascular diagnosis device using ML

3rd Place (12 teams) — InBody Hackathon 2023 (\$1450)

Jul 2023 – Jul 2023

• Autonomous mobile robot for executing missions

3rd Place (20 teams) — AI Object Detection-Based Solar Tracking Robot Challenge (\$220)

Jul 2023 – Jul 2023

• Dating app "Sidaeting"

1,700 User Base

Implemented a matching algorithm that combinatorially optimizes collective happiness from users' ideal type data and developed the frontend for "Sidaeting", a dating service launched by UOSLIFE.

Mar 2022 – Jun 2022

SKILLS

• **Languages:** Korean, English

• **Technical Languages:** C, C++, Java, JavaScript, Julia, Python, SQL, HTML, CSS

• **Technical Frameworks:** ROS1-2 (Melodic, Noetic, Foxy, Humble, Jazzy), MuJoCo, PyTorch